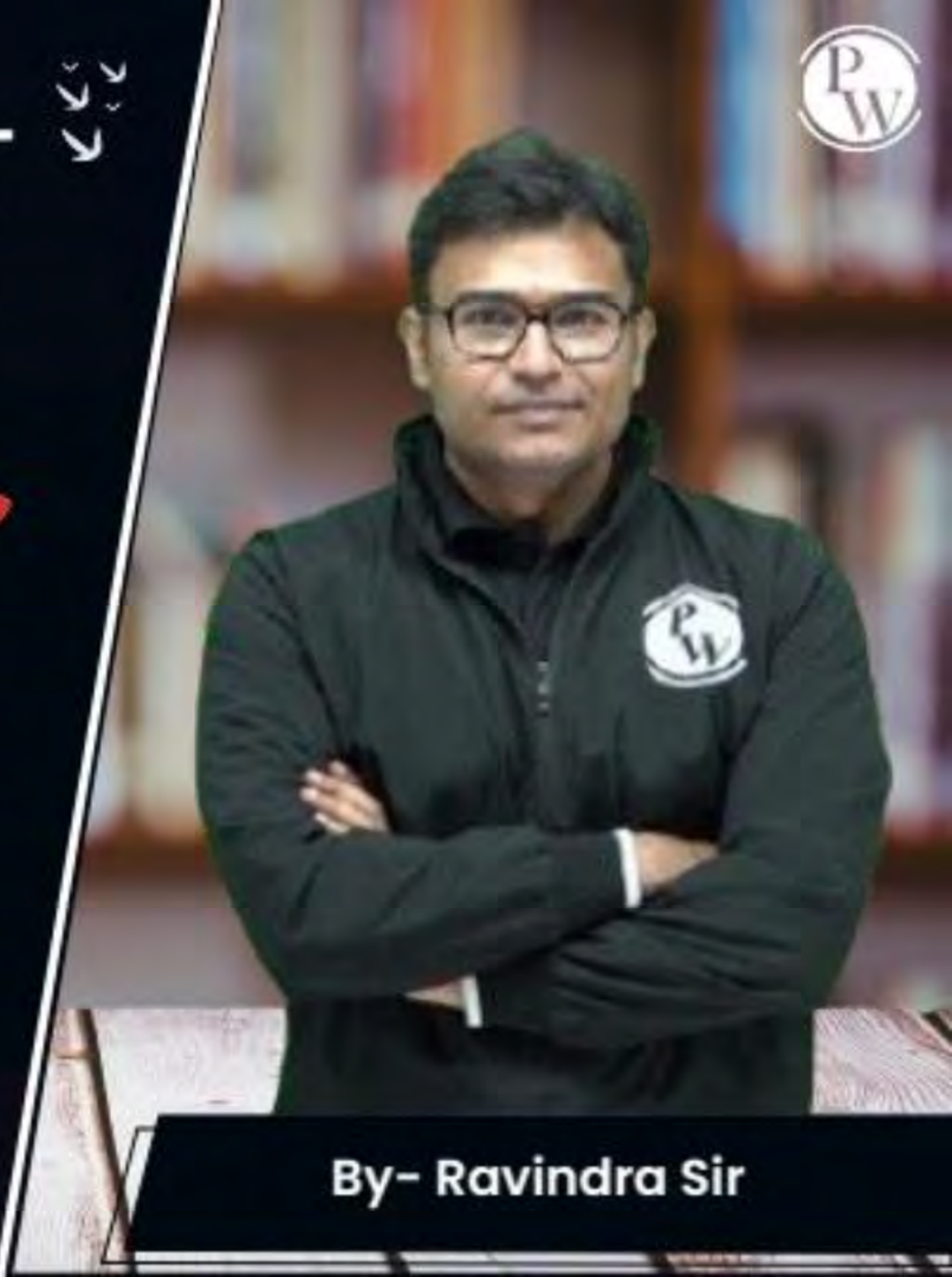
Computer Science & IT

Theory of Computation



Regular Languages

Lecture No. 10



By- Ravindra Sir

Recap of Previous Lecture



Topic NFA

Topic DFA

Topic RE

Topics to be Covered



Topic NFA

Topic DFA

Topic RE

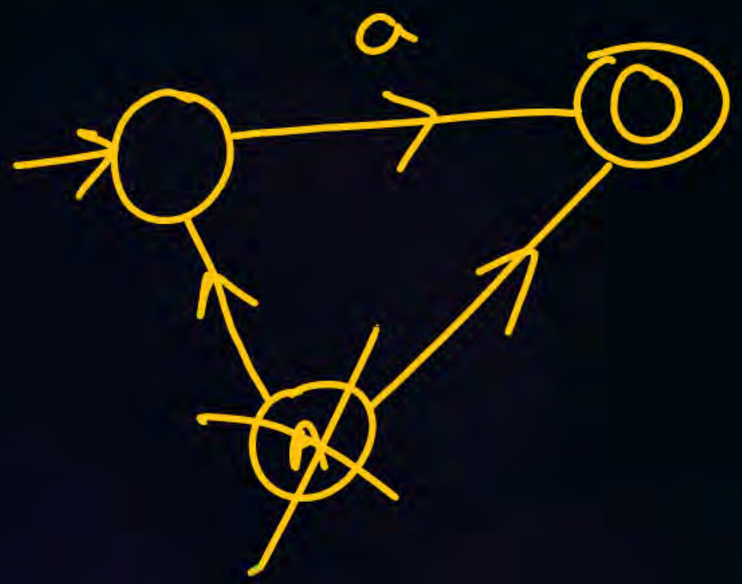
Finite automata → Regular exp

1) intuition → FA → language → R.E.
2) State elimination (Fast and easy)
3) Ardens theorem (slow)

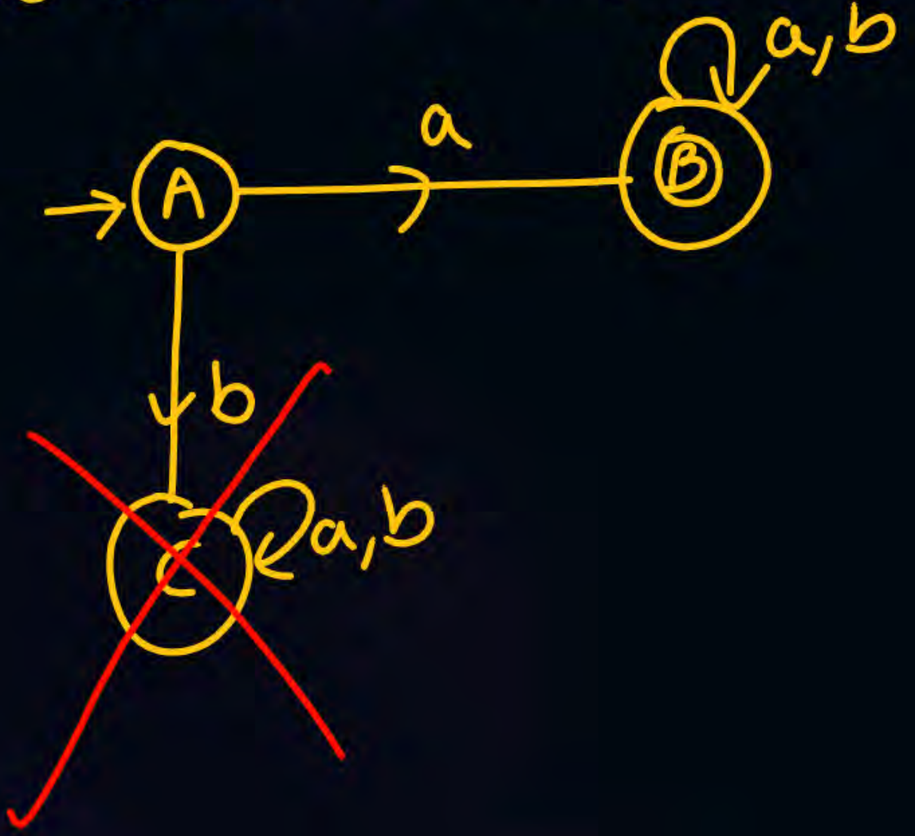
State elimination method :-

→ It can be applied on DFA, NFA, ϵ-NFA

→ Remove any unreachable states

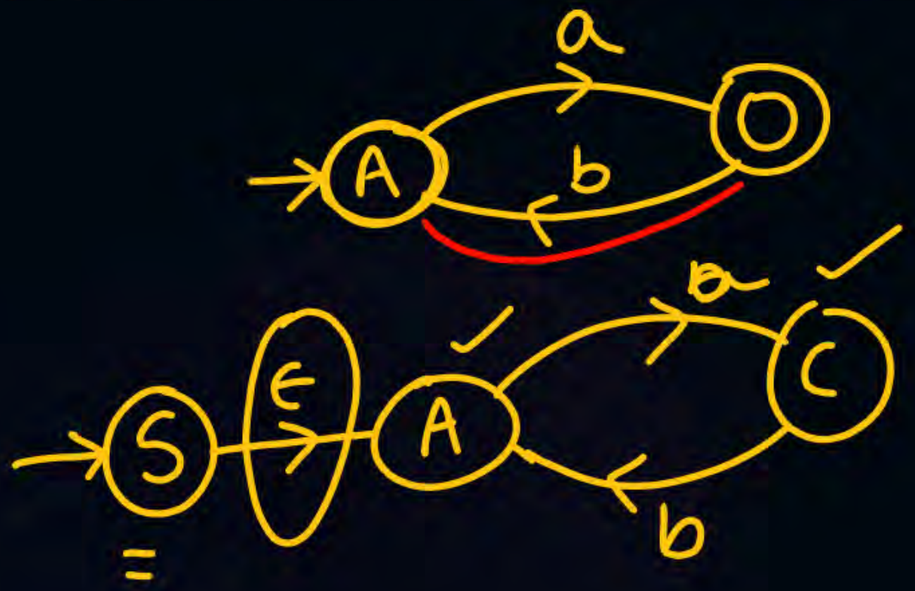


→ Remove dead & trap states



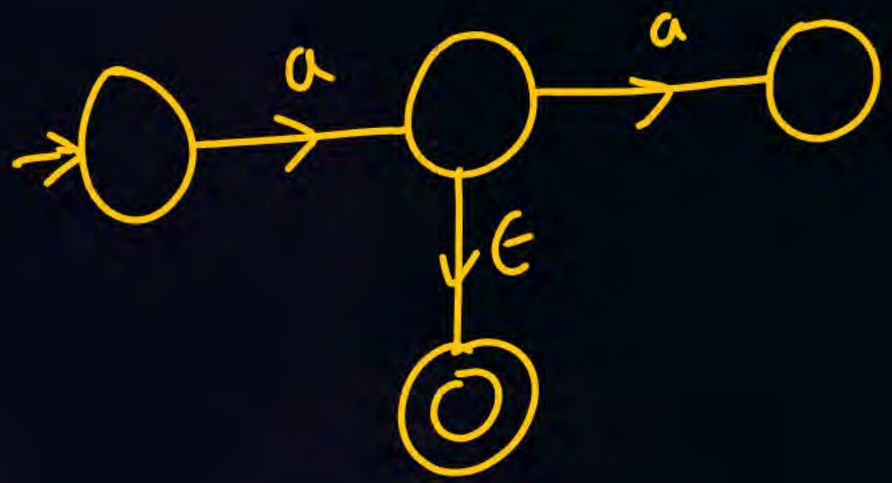
→ merge the final states if possible

→ There should not be any incoming edges into initial state



→ There should not be any outgoing edges from final state

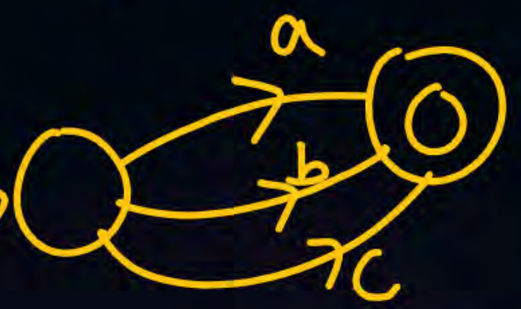




→ There should be only one final state



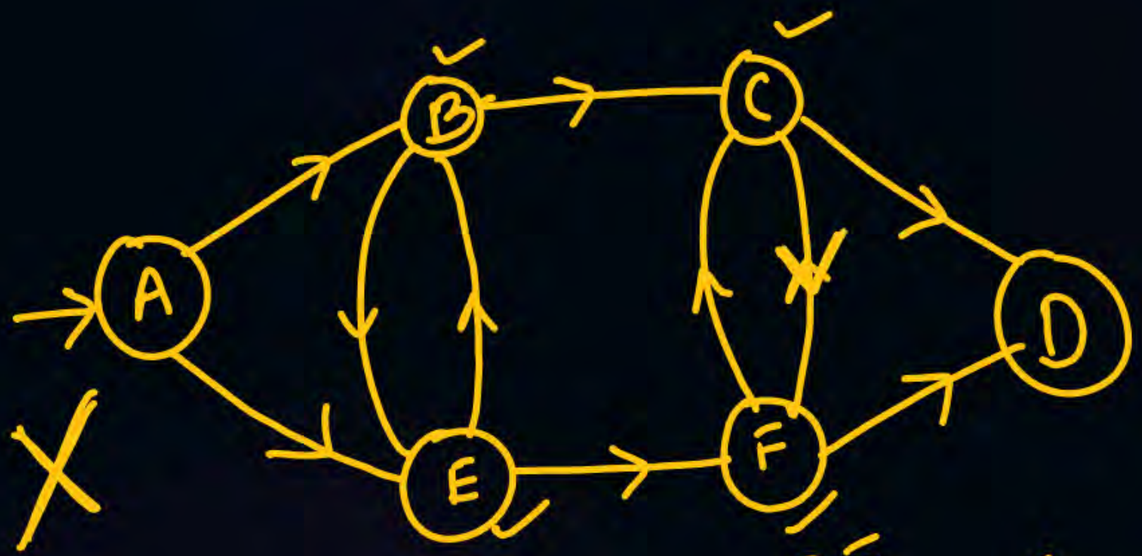
→ Eliminate the states in any order except initial and final states

Ex- 

RE: a+b+c



ab



BCEF — RE

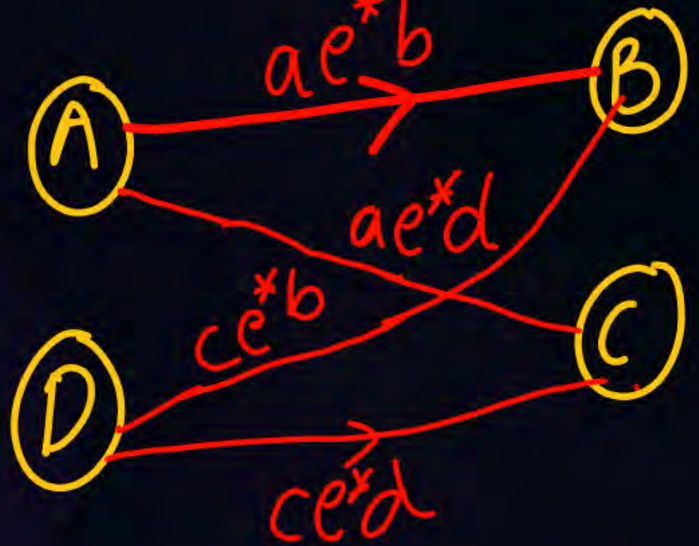
BCFE — RE

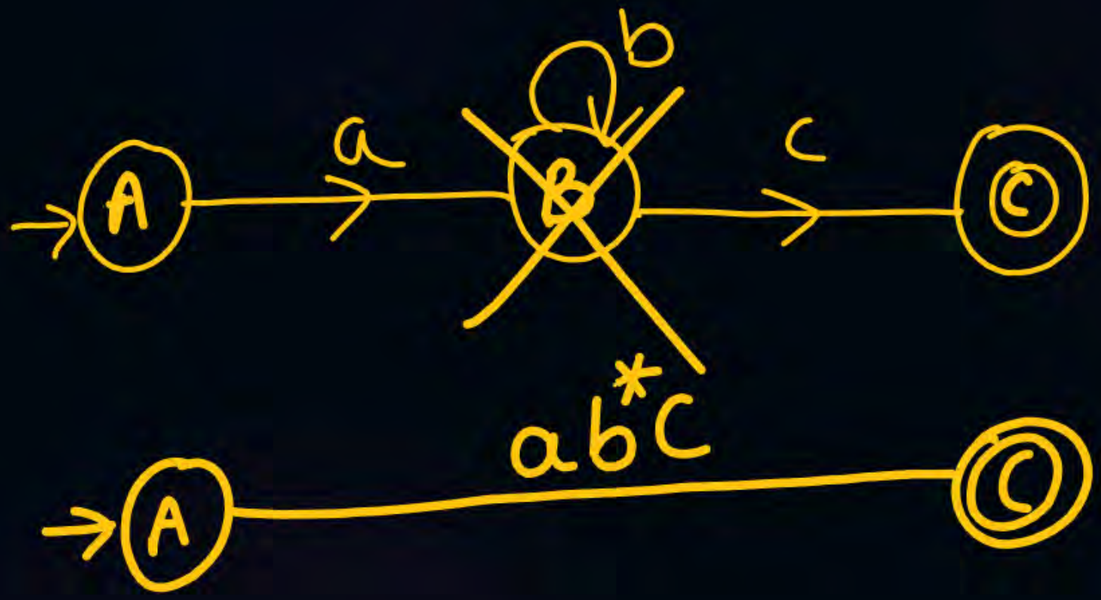
CBFE — RE

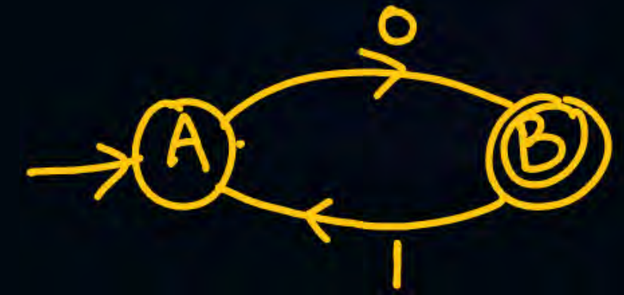
4! = 16 → RE



E → incoming = 2
Out going = 2




A
a
B
b
c
C
A
ab*c
C

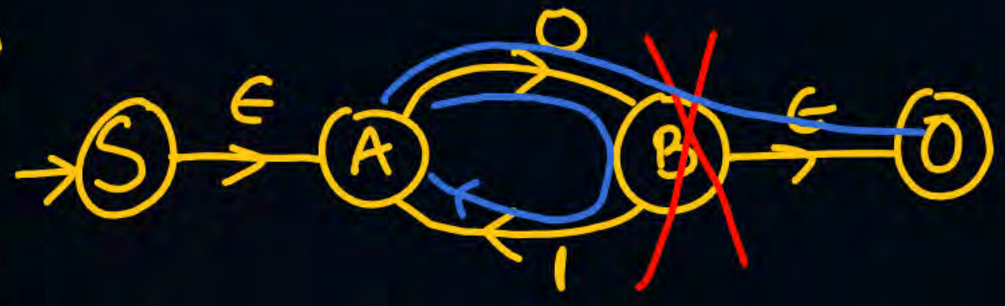




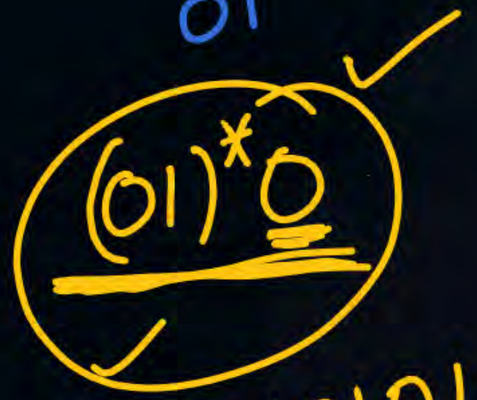
A - incoming → 2
A - outgoing → 1

$2 \times 1 = 2$

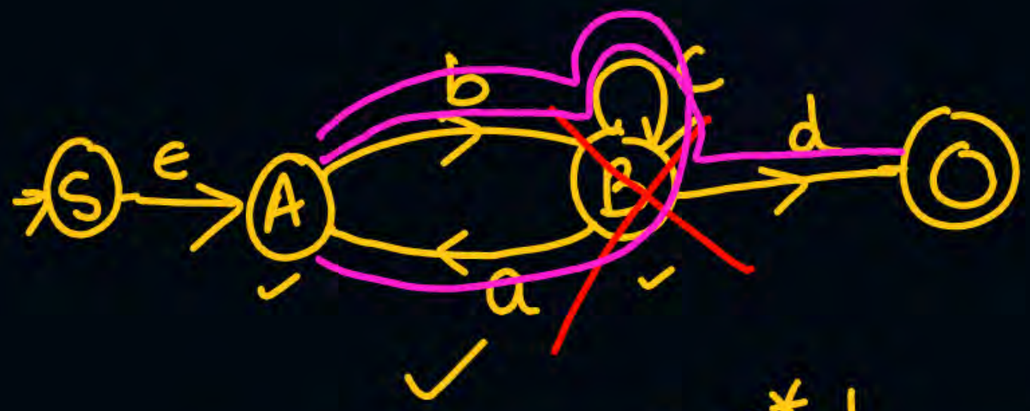


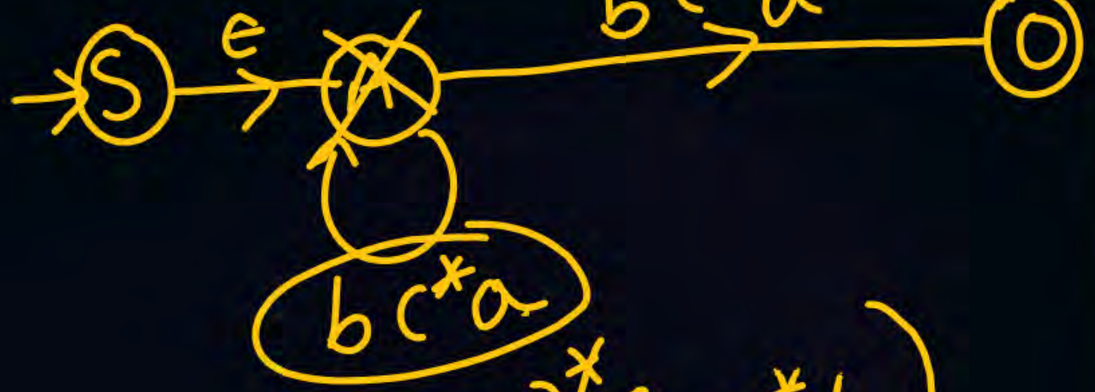






0101010



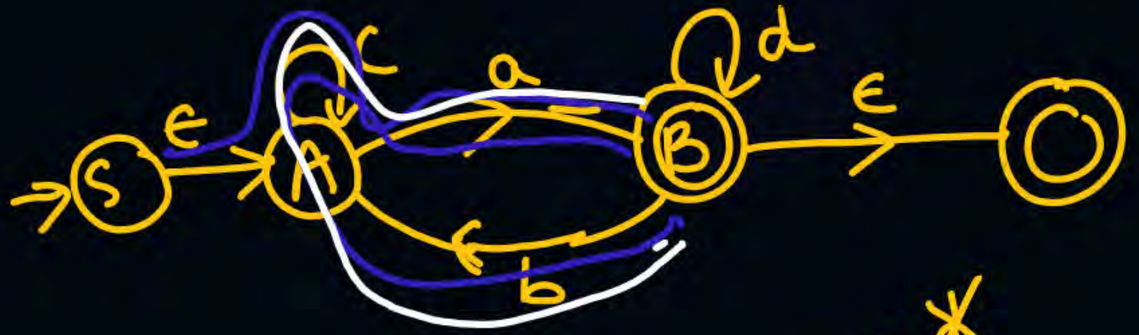


$(bc^*a)^*(bc^*d)$

Hw

A → B = II

A → I

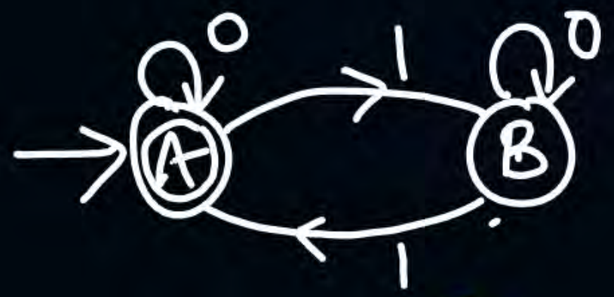


HW
Eliminate B, then A



$c^*a(bc^*a+d)^*$



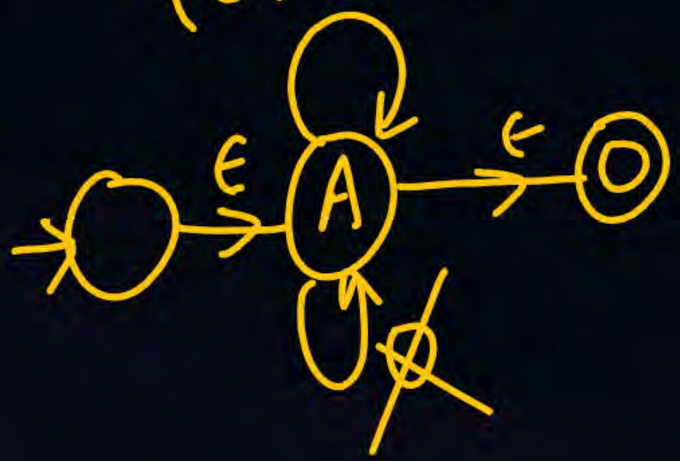


→ even no of one's



$(0+10^*1)^*$

$(0+10^*1)$




A
B
C
0
1
2, 3
S
ε
10*10*1+0
*
2 num

Ardens lemma

$S = R + ST$

$S = RT^*$

RHS: $R + \underline{S}T$

$R + RT^*T$

$R(\epsilon + T^*T)$

$= RT^*$

$S = R + TS$

$S = T^*R$

X use adem

RHS $R + T\underline{S}$

$R + TT^*R$

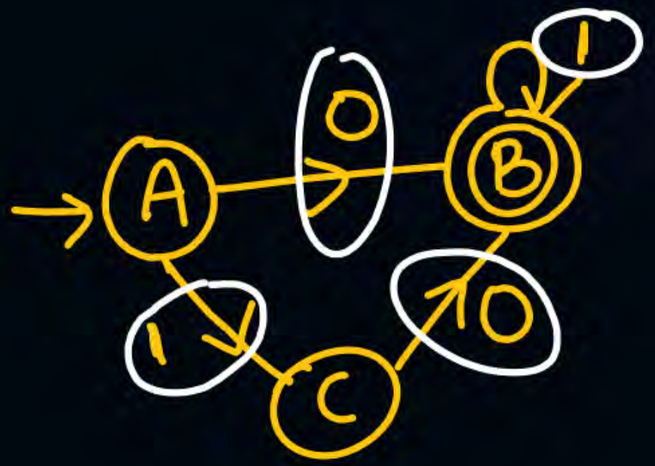
$(\epsilon + TT^*)R$

T^*R





$(0+10)(1)^*$



$=$

L

$C = A1$



$A = \epsilon \rightarrow (1)$

$B = A0 + C0 + B1 \rightarrow (2)$

$C = A1 \rightarrow (3)$

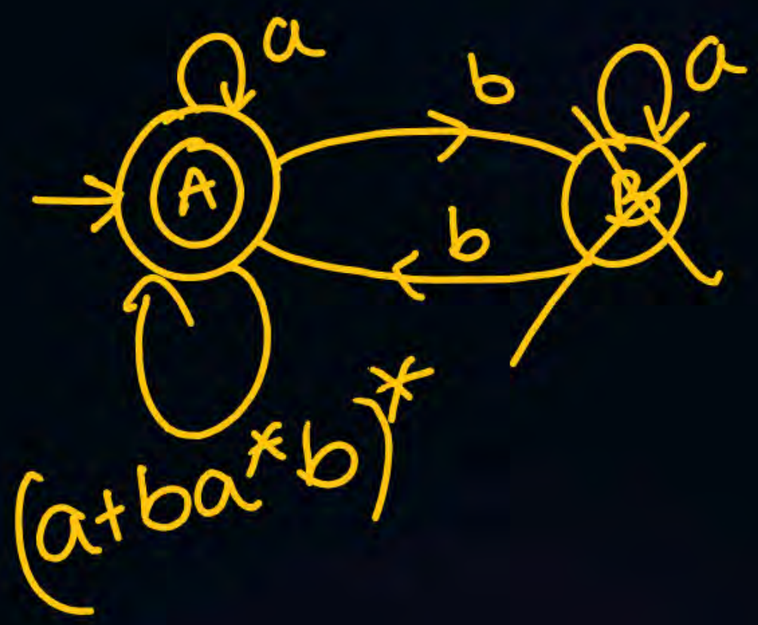
$C = \epsilon 1$

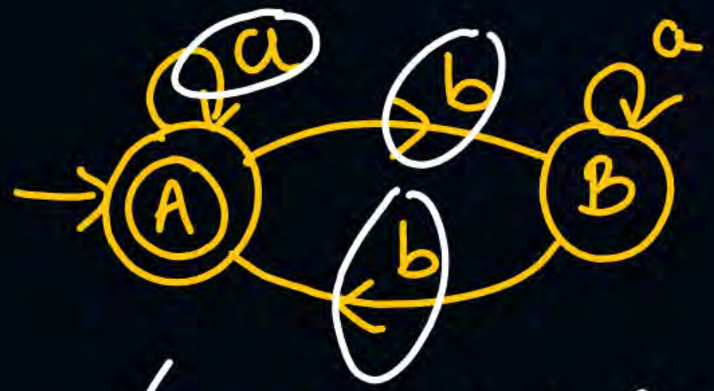
$C = 1$

$\frac{B}{S} = \frac{0+10}{R} + \frac{B}{S}\frac{1}{T}$

$S = RT^*$

$B = (0+10)1^*$


a
b
a
A
B
b
(a+ba*b)*



$A = \epsilon + Bb + Aa$

$B = Ab + Ba$

$S = RT^*$

$B = Aba^*$

$A = \epsilon + Aba^*b + Aa$

$A = \epsilon + A(ba^*b + a)$

$A = \epsilon(a + ba^*b)^*$

A is final state

$A = 10 + 01A + B$

$B = 00 + CB$

$\frac{C}{S} = \frac{11}{R} + \frac{0C}{T\ S}$

$S = T^*R$

$C = 0^*11$

$\frac{B}{S} = \frac{00}{R} + \frac{0^*11}{T}\ \frac{B}{S}$

$S = T^*R$

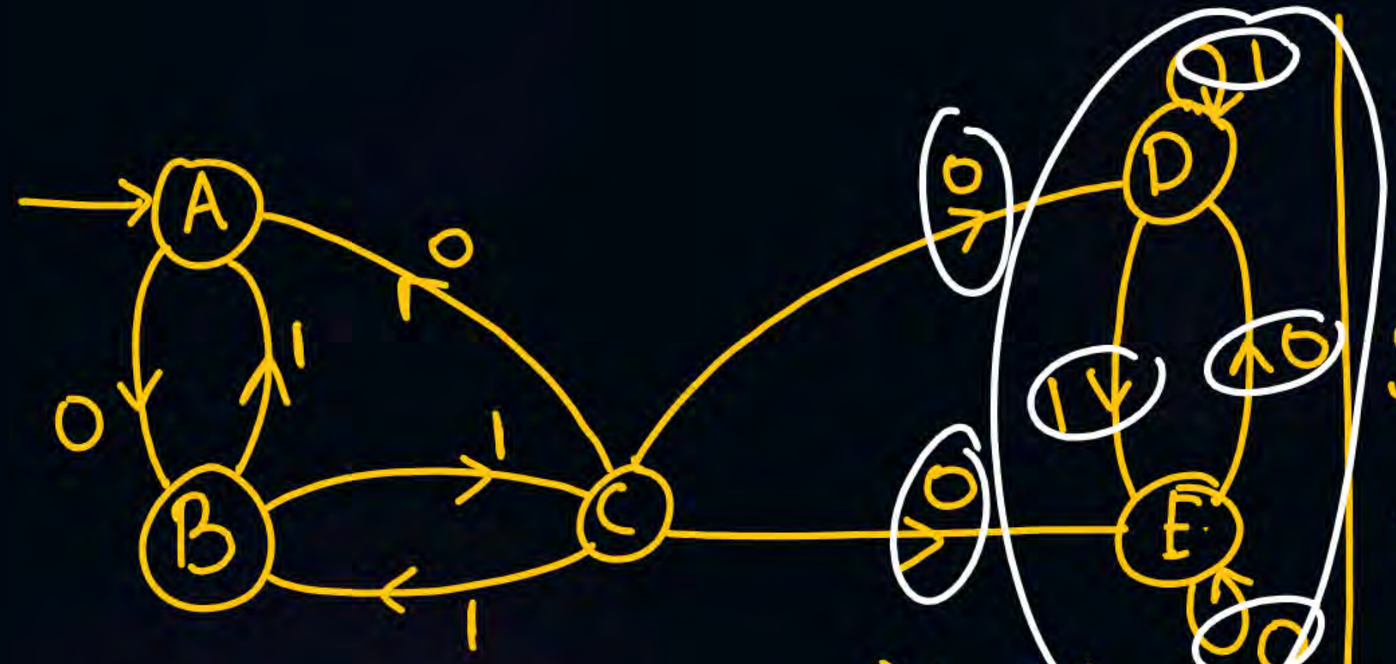
$B = (0^*11)^*00$

$A = 10 + 01A + (0^*11)^*00$

$\frac{A}{S} = \frac{10 + (0^*11)^*00}{R} + \frac{01}{T}\ \frac{A}{S}$

$S = T^*R$

$A = (01)^*($ $\because (0^*11)^*00$)



If D is final state then language is L_D

If E is final state then language is L_E

a) $L_D = L_E$ b) $L_D \subseteq L_E$ c) $L_E \subseteq L_D$

d) $L_D \neq L_E$

$D = C0 + D1 + E0$

$E = E0 + B1 + C0$

DFA, NFA, RE

Grammars :- (V, T, P, S)

LHS → Vertices

Terminals

Productions

Start Symbol

$S \to a$ → production

$S \Rightarrow a$ → derivation



→ derivation tree

$S \to a$
$S \to b$

$S \to a / b$

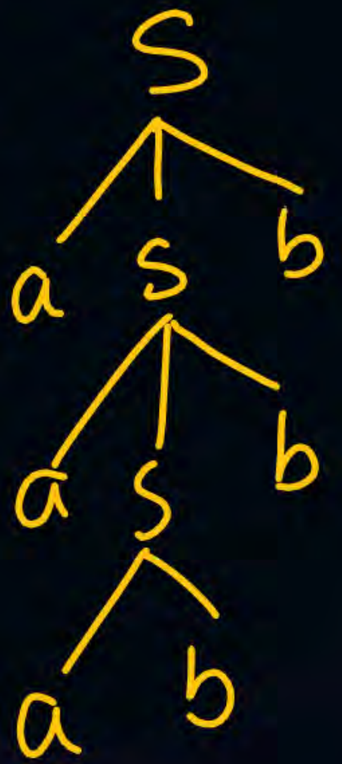
$S \rightarrow aSb / ab$

$S \Rightarrow aSb \Rightarrow aabb$ → derivation



yield

aabb



aaabbb

Grammars

T0

T1

T2

✓ T3 → Regular grammars

Regular grammars

Left linear grammar

$A \rightarrow A\alpha / \beta$

$\alpha, \beta \in T^*(\Sigma^*)$

Right linear grammar

$A \rightarrow \alpha A / \beta$

$\alpha, \beta \in T^*(\Sigma^*)$

$S \to aS/bS/\epsilon \to$ RLG

$S \to aaaS/ab \to$ RLG

$S \to Sa/Sb/ab \to$ LLG

$S \to Sa/ab \to$ Not reg X

XLL

XRL

$S \rightarrow aS/Sb/aa \rightarrow$ not reg

$S \rightarrow SS/a/\epsilon \rightarrow$ not reg

$L = \{\epsilon\}$

$S \rightarrow \epsilon \rightarrow$ Reg

$L = \{a\}$

$S \rightarrow a \rightarrow$ Reg

$L = \{a, b\}$

$S \rightarrow a$
$S \rightarrow b$

$S \rightarrow a/b$

$L = a^*$

$S \to aS/\epsilon \to$ Reg → RLG

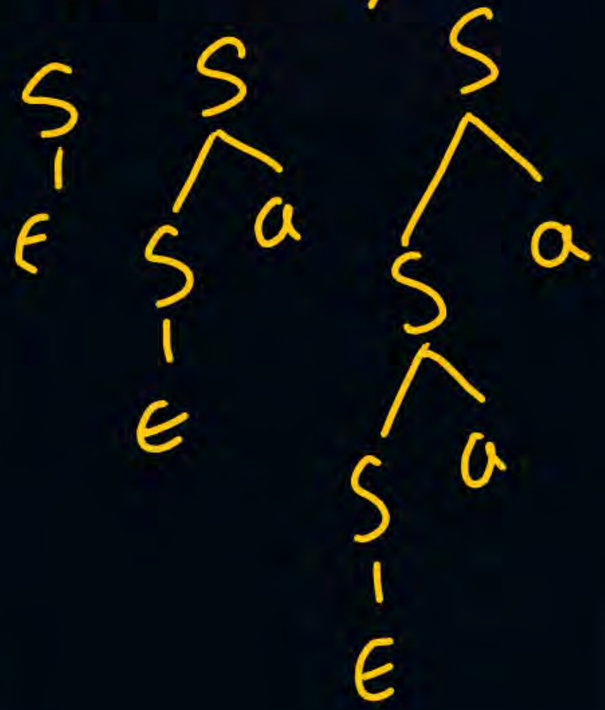


$S \to SS/a/\epsilon \to a^*$

CFG

not regular

$S \to Sa/\epsilon \to$ LLG → Reg grammar

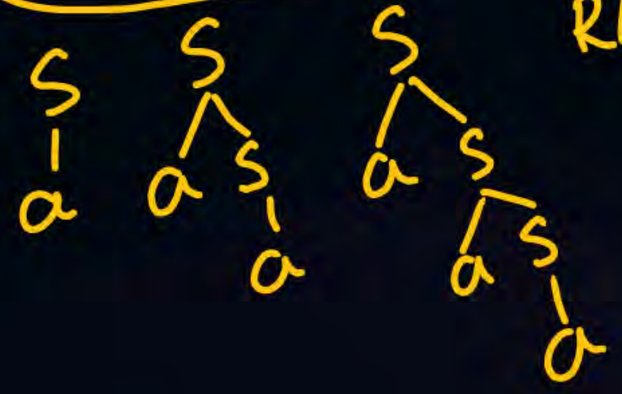


All Regular grammars generate Regular languages

But Some non regular grammars also generate Regular languages

$L = a^+$

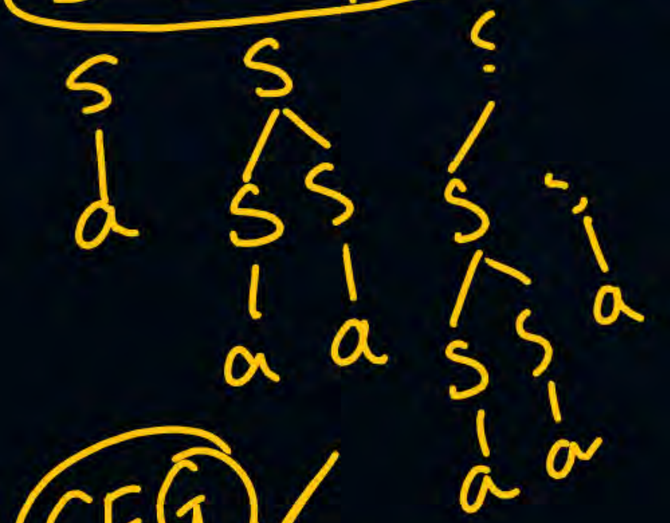
$S \to aS \mid a$ → Reg → RLG



$S \to Sa \mid a$ → Reg → LLG



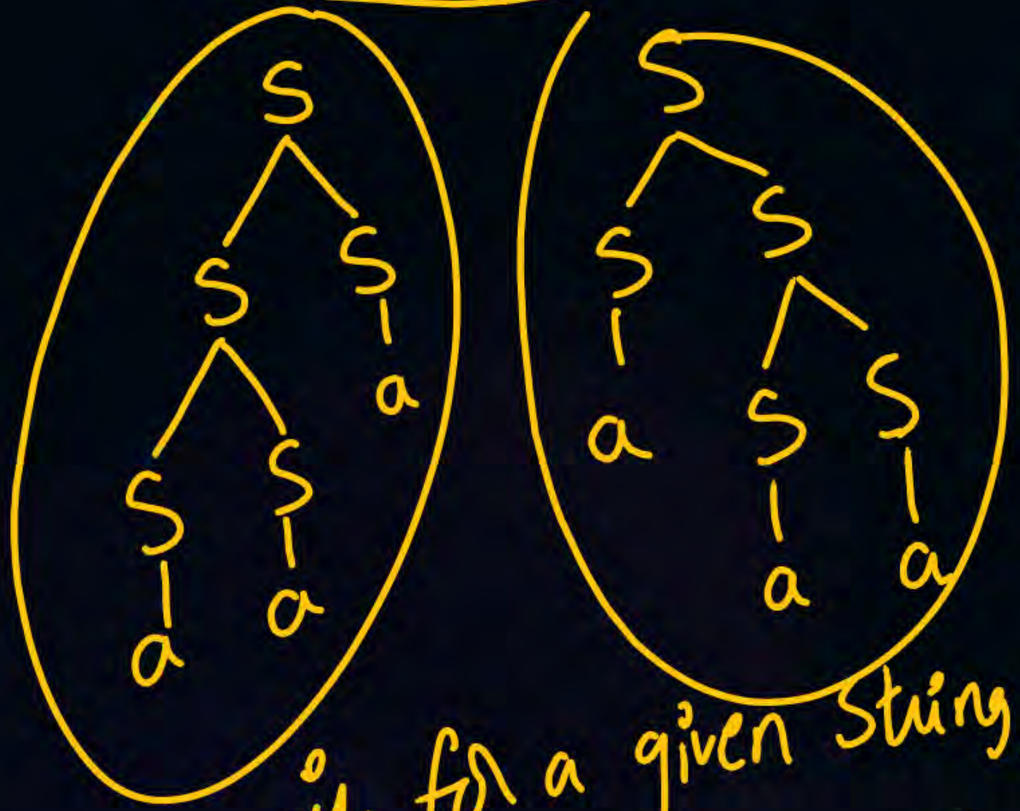
$S \to SS \mid a$ ✗ Reg



CFG ✓

a^+ → Reg

$S \to SS \mid a$ → ambiguous



CFG

RMD
LMD

Compiler

if for a given string, more than one parse tree is possible then the grammar is called ambiguous.

$(ab)^*$

x^*

$S \to xS/\epsilon$

$S \to abS/\epsilon$

↓

Reg

RLG

$S \to Sx/\epsilon$

$S \to Sab/\epsilon$

↓

Reg

LLG

$(1101)^*$ → RLG

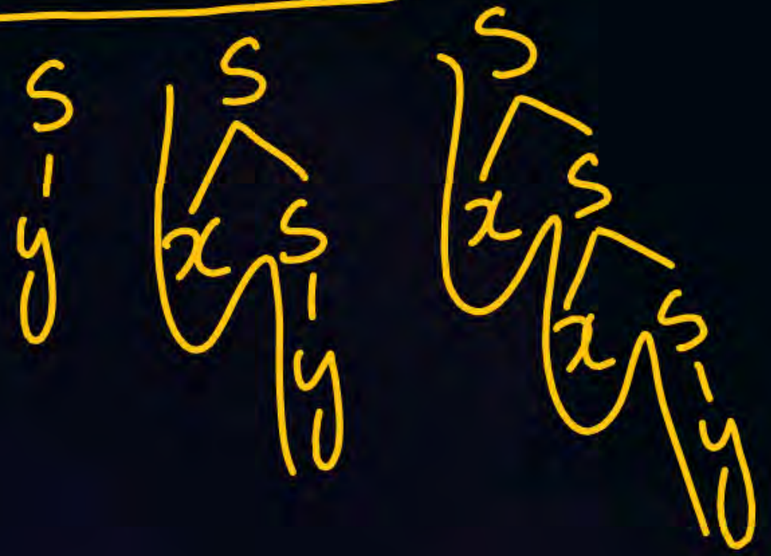
$S \to 1101S/\epsilon$

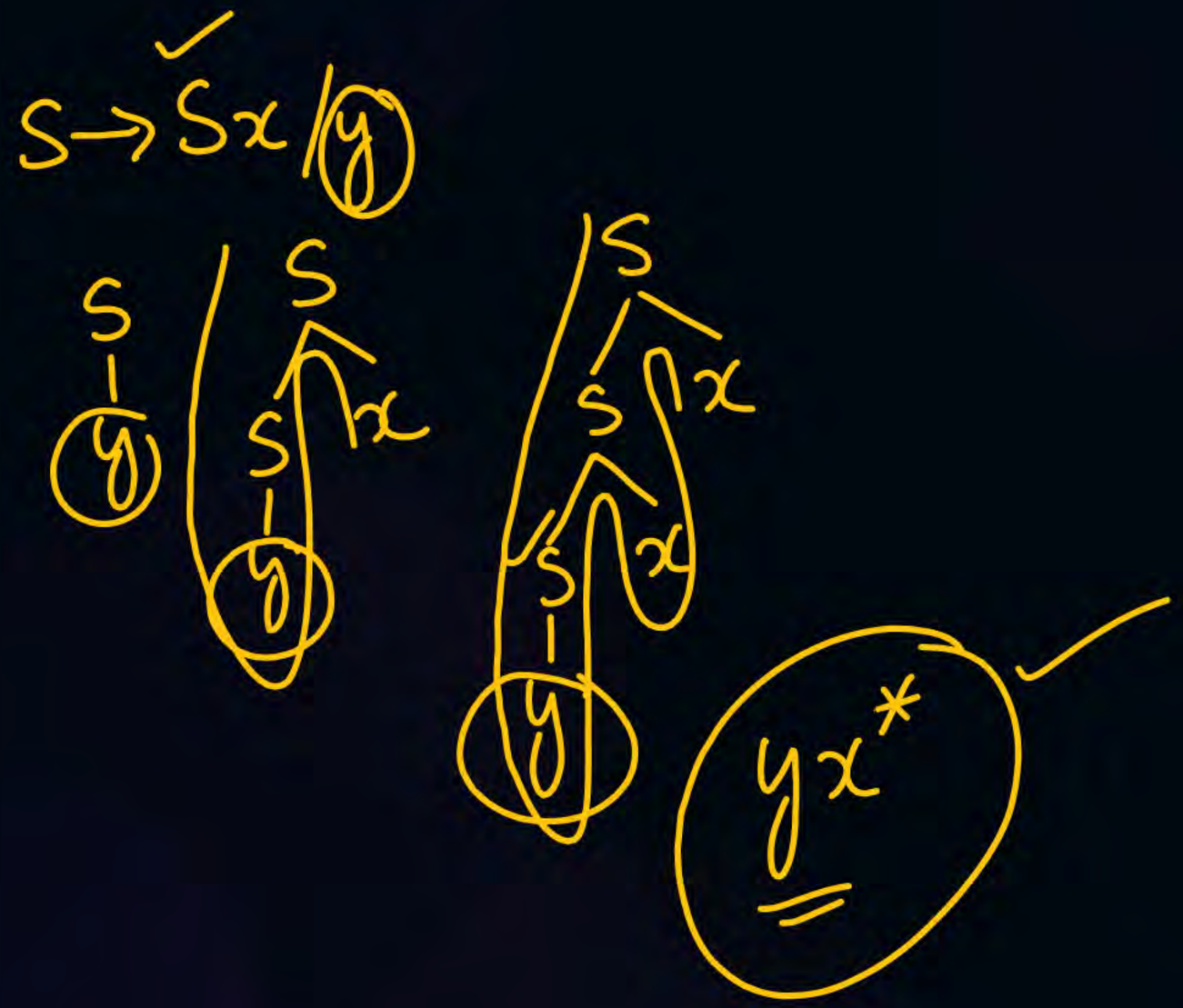
$S \to S1101/\epsilon$ → LLG

→ Reg

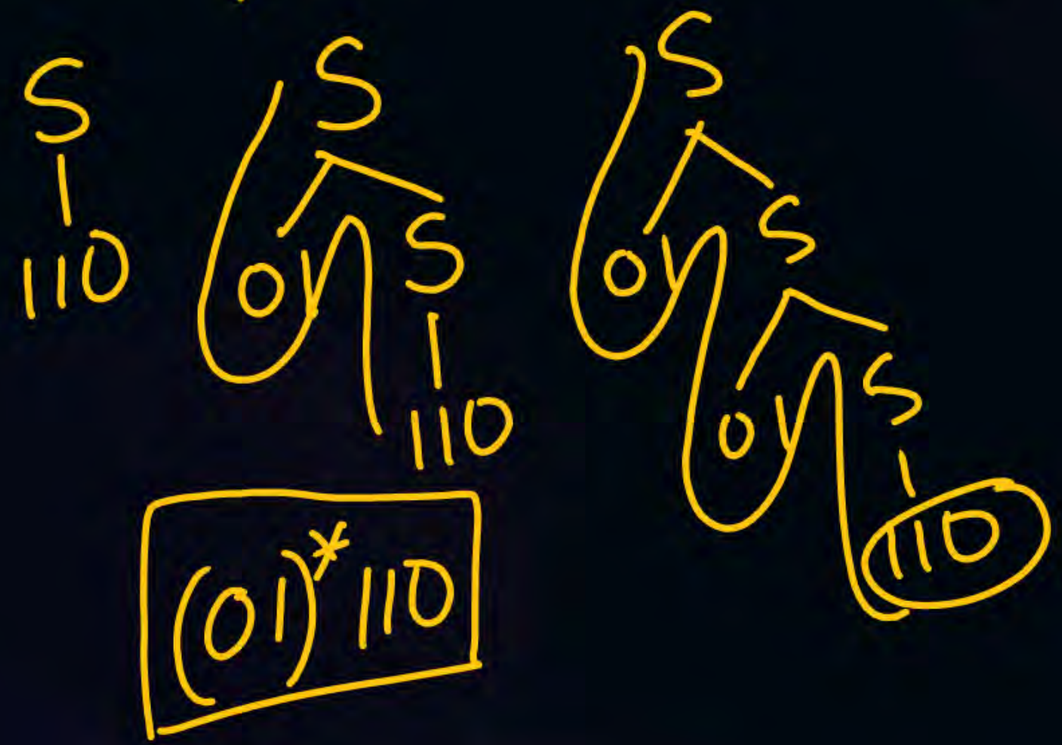
$\rightarrow x^* y = \{ y, xy, xxy, xxxy \cdots \}$

$S \rightarrow xS / y$




S → Sx / y
S
y
S
S
x
y
S
S
x
S
x
y
yx^*

$S \to 01S / 110 \to RLG \to Reg$



$(01)^* 110$

THANK - YOU